SAMSUNG PRISM GENAI HACKATHON 2026-27 · THEME 05 · ROUND 2 · LIVE DEMO

DUET for Galaxy

Interrupt it, hesitate, change your mind.
It acts on what you finally meant: once, and only says what happened.

TWO SPEEDS · ONE COORDINATOR · REAL ACTIONS

ON A GALAXY, TODAY

0 · 0	stale · repeated actions in 36 spoken test conversations
0.9 s	to “Sure.” after you stop talking (p50)
0	wrong actions by the fast mind in 62 decisions
0.81	Round 1: FDB-v3 Pass@1 (GPT-Realtime 0.60)

01 HEAR — words, while it talks → **02 ROUTE** — fast mind or cloud mind → **03 WAIT** — until you're done → **04 ACT ONCE** — never on words you took back → **05 CONFIRM** — from the device's own reply

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Gemma 4 E2B on our GPU · Gemma 4 26B in the cloud · LiveKit · Whisper · Kokoro · SmartThings · Android

Voice assistants act on the first thing you said

People correct themselves mid-sentence. An agent that acts at the end of every phrase does the wrong thing, or both things.

"Set an alarm for 6:30… no, make it 7."

A greedy assistant

`set_alarm(6:30)` then `set_alarm(7:00)`

two alarms: you wake at 6:30

A cautious one

`waits for you to confirm`

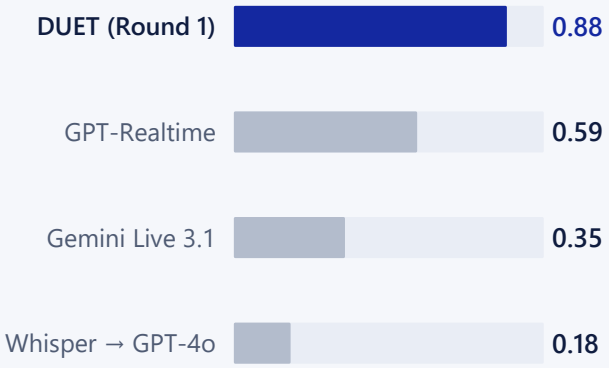
slow, and asks what you already said

DUET

`6:30 never runs · 7:00 runs once`

"Sure. Alarm set for 7 in the morning." ~1.5 s

SELF-CORRECTIONS, FULL-DUPLEX-BENCH V3



Strict Pass@1 on the benchmark's self-correction items. Baselines from the FDB-v3 paper: "models commit intermediate parameters before the correction arrives."

WHO NEEDS IT

Anyone whose hands or eyes are busy: driving, cooking, an older person living alone.

WHAT GOES WRONG TODAY

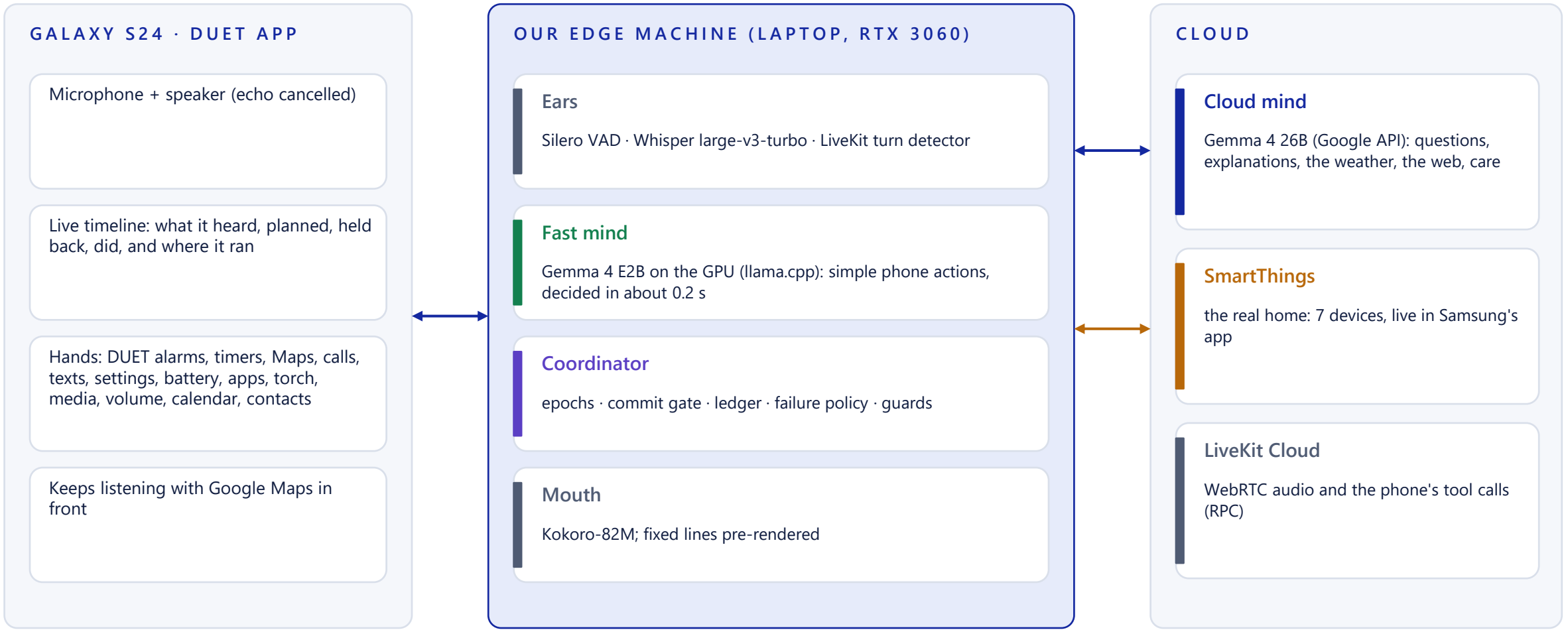
Stale actions, duplicate actions, claims of "done" that did not happen, silence while it thinks.

WHAT DUET CHANGES

Acting is gated by deterministic code, not by the model's timing. The model can be wrong about when; the user never pays for it.

The phone is the ears, mouth and hands; two minds share one coordinator

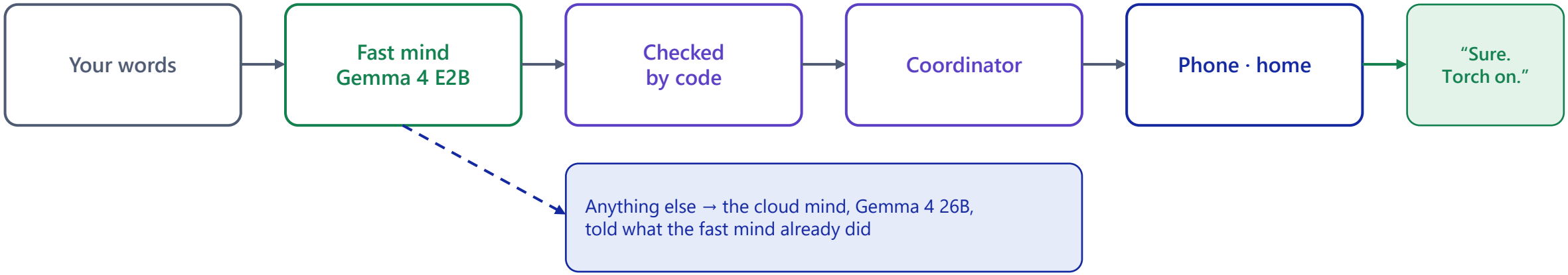
What runs where in today's demo. Every arrow into the phone or the home passes the coordinator first.



Simple actions never wait for the cloud; questions never wait for the phone. Models are configuration; the coordinator is the product.

A small Gemma acts in about a second, and hands over anything unclear

An on-device and cloud split, as today's assistants use, with one guarantee for both: code checks every plan before it runs.



THE FAST MIND MAY ACT ONLY WHEN

✓ **Named in your words**	“open the shopping apps” is not open_app(apps); “dentist at 4” is not an alarm
✓ **Values you said**	a contact, a place, a number the user never said goes to the cloud mind
✓ **The last value said**	“3:30, no, cancel that, 430 instead” must plan 4:30
✓ **Not a question**	“Is the AC on?” is answered, never switched on
✓ **Nothing dropped**	“…actually never mind” is the cloud mind's to judge

THE FAST MIND ALONE

0

wrong actions in 62 decisions

0.23 s

to decide (p50)

Held-out sentences: 14/14. A plan that fails a check costs a second, never a wrong action.

Plan early, commit late, act once: whichever mind made the plan

Deterministic, about 250 lines, covered by unit tests. The same code passed Round 1's benchmark.

Epochs

Every new word voids older plans, even after the turn has closed. A cough only delays them.

Commit gate

An action waits until you have been quiet 1.4 s on the phone (1.8 s after “no wait”, 2.2 s after an open sentence). Thinking starts at once; only acting waits.

Idempotency ledger

The same action never runs twice, even if you barge in while it runs.

Failure policy

A failed reading is retried once; a failed or unconfirmed change is never re-sent, and you are told.

GUARDS ADDED FOR THE PHONE: THEY HOLD WHATEVER A MODEL DECIDES

“Call my…”	cut off mid-phrase: nothing is done; DUET asks for the rest
“6:30, or maybe 7, hmm”	undecided: nothing is done; DUET asks which
“Set an alarm for 6:30” at 11:50 pm	a time without AM/PM means the next one: 6:30 tomorrow morning
“Turn off the light” (two lights)	a name that fits two devices is asked about, never guessed
A request the user dropped	never acted on by the fast mind; the cloud mind judges what is left

Android's public interfaces, a real SmartThings home, and the world

Three modes: Assistant for every Galaxy owner, Care for an older person living alone, Drive for eyes on the road.

Alarms	DUET's own: exact set, cancel, list; ring; on the lock screen	REAL
Battery & heat	charge, drain, temperature, health, screen, apps used	REAL
Apps & settings	open, count, the right settings screen	REAL
Calls & texts	your contacts; direct or with one tap	REAL
Calendar	events in the phone's calendar	REAL
Weather	Open-Meteo, any town, the next 6 days	REAL
Medicines	schedule and double-dose guard (Care)	ON-PHONE DATA
Timers	the Clock app	REAL
Brightness & timeout	changed on request, never unasked	REAL
Maps	navigation; DUET keeps listening behind it	REAL
Torch · media · volume	on, off, play, pause, skip, set	REAL
SmartThings	lights, AC, TV, lock, vacuum, washer: live in Samsung's app	REAL API
Web answers	Gemma with Google Search grounding, with sources	REAL
The car	trip and climate (Drive)	SIMULATED, LABELLED

It never says something is done before the device says so

Each rule below exists because a test caught DUET, or Android, doing otherwise.

Confirmed from the device's reply

“Alarm set for 7 in the morning.” is built from Android's own answer, with no model call, the moment the action is done.

Sent is not done

A SmartThings change counts only when the device reports its new state; otherwise DUET says it was sent, not done.

Unknown stays unknown

If the phone never answers, the action is never re-sent and DUET says it may not have gone through. (A test caught “I've set a timer” here.)

No cloud? Said at once

After 2 s of not reaching Google, DUET says so; simple actions keep working on the fast mind. A drill button cuts the cloud live.

Android's limits, said plainly

Behind Google Maps, Android blocks opening other apps unless DUET may appear on top: DUET says what to allow instead of pretending.

Its own alarms

Google's Clock ignored a cancel request and Android lets no app check the Clock's alarms. So DUET keeps exact alarms of its own.

Four kinds of tests, run again after every fix

Spoken scenarios have three kinds of expectations: what DUET must do, must never do, and must never claim.

139

unit tests

every rule: the gate, the ledger, the guards, the routing, the key pool

36

spoken scenarios

the Galaxy Correction Suite: corrections, wavering, barge-in, noise, drills

19

end-to-end cases

the real app on Android 16, also behind Google Maps and with the cloud cut

62

fast-mind decisions

with held-out sentences written after its checks

THE LATEST FULL RUN

34/36

spoken conversations passed

0 · 0

stale · duplicate actions

Fast mind on our GPU, cloud mind Gemma 4 26B, a simulated phone answering exactly as the app does, the real SmartThings home.

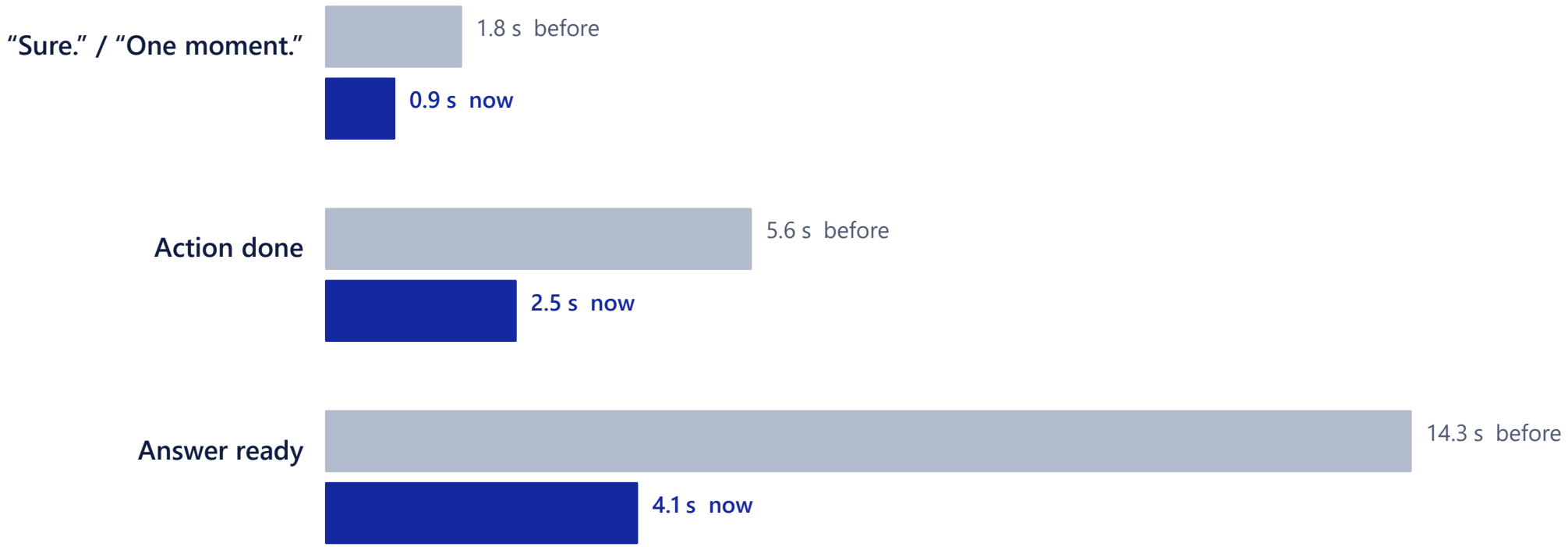
The two misses (a misheard time, a light turned on that was already on) were fixed and re-run: 10 of 10.

BUGS OUR TESTS CAUGHT BEFORE ANY JURY DID

- ✗→✓ “Garage door” matched the front door lock
- ✗→✓ “Turn on the AC at 22” left it at 24
- ✗→✓ A timer the phone never confirmed was later called “set”
- ✗→✓ A blocked call was retried eight times
- ✗→✓ “Call my” … “my daughter” lost the word “call”
- ✗→✓ A dropped alarm was set on a second look
- ✗→✓ The Clock app ignored a cancel; Android blocked actions behind Maps

No dead air: something useful within a second

Seconds from the end of the user's speech, p50. Before: the phone app on 30 Sep (cloud mind only). Now: two speeds, the latest suite.



Fast mind first

simple actions never wait for the cloud

Minimal thinking

the cloud mind answers without a long reasoning step

Acknowledged by route

“Sure.” the moment the fast mind acts; “One moment.” the moment the cloud takes over

Confirmed from the device

no second model call to say “done”

The same split moves onto the phone; the coordinator moves with it

Today the brain runs on our edge machine and Google's cloud. The design was made to fit on the device.

Today

Laptop edge: ears, fast mind, coordinator. Phone: the hands. Cloud: depth.

Next: on the phone

Kotlin coordinator (a 1:1 port, the same tests); Gemma 4 E2B on-device with Google's LiteRT-LM; on-device speech; the cloud only for depth

With Samsung

Bind to Bixby's device agents, AppFunctions and SmartThings directly; Samsung Clock's own API; the side button

What it is not, yet

The phone app's brain runs on a laptop today (the phone is ears, mouth and hands). The car is simulated. English only.
No payments, no acting inside other apps (Android's rules), no emergency calls: it says so.

Why it matters for Samsung

Any assistant that acts (Bixby's agents, Car-to-Home, an AI contact centre) needs the same guarantees: never act on a corrected value, never twice, never claim before it happened.
DUET is that layer: models are configuration, the coordinator is the product.

Thank you. Now please interrupt it yourselves.